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Power Generation Fuel Clustering Analysis Using K-Means

Executive Summary

This study used K-means clustering to analyze monthly fuel receipt and cost data from the EIA-923 dataset to discover patterns in fuel usage among U.S. power generation facilities. After cleaning and standardizing the data, five clusters were identified based on fuel quantity, energy content, fuel cost, sulfur content, and ash content. The results showed distinct groups of facilities that differed in operating scale, fuel quality, environmental characteristics, and fuel costs. The clustering was also closely aligned with fuel types, even though fuel type was left out of the clustering model, indicating that the selected numerical variables appropriately captured meaningful operational differences. One cluster represented extreme fuel price outliers, demonstrating K-means' sensitivity to unusually large values. Overall, the analysis finds that power plants can be effectively segmented by fuel consumption patterns, giving awareness of operational characteristics and environmental impacts.

In addition to demonstrating that U.S. power plants can be meaningfully clustered using this dataset, the analysis also provided useful insights for future energy strategies in the United States. The underlying data, discussed in detail later in this paper, revealed a strong correlation between fuel source and pollutant emissions, with coal usage associated with the highest pollutant levels. However, clusters dominated by coal products also exhibited the highest energy performance per unit. These results indicate that, while transitioning to cleaner and more sustainable fuels such as natural gas is desirable, it may continue to be necessary to supplement the energy supply with higher-emission fuels like coal until alternative fuel processing becomes more efficient.

Introduction

The dataset used in this project was the monthly fuel receipts and cost information from the U.S. Energy Information Administration (EIA-923 Schedule 2, Part A). The dataset contains detailed information on fuel purchases for electric power plants, including fuel quantities, energy content, fuel costs, sulfur content, and ash content.

Several preprocessing steps were completed prior to the clustering analysis. To reduce processing requirements, the original dataset of nearly 300,000 records was sampled to a controlled subset representing 2 percent of the data. Text values of "null" were converted to missing values, and variables with more than 33% missing observations were removed. Remaining incomplete observations were omitted using listwise deletion. Variables such as plant names, utilities, states, and fuel types were converted to categorical variables, while fuel cost variables were converted to numeric values. Identifier variables were converted to character variables to prevent them from influencing the clustering process. Because the numerical variables were measured on different scales, all clustering variables were standardized as z-scores before applying K-means clustering. After data cleaning and normalization, variables for use in the K-means clustering algorithm were selected. Since K-means clustering relies on Euclidean distances, only the remaining numerical variables were used for cluster analysis. The fuel type code was incorporated after clustering to describe the data and identify patterns related to fuel type.

Problem Statement

The objective of this project was to determine whether U.S. power plants could be grouped into meaningful clusters based on fuel receipt, emissions, and cost characteristics. Specifically, this analysis sought to answer the following questions:

- Can K-means clustering recognize distinct groups of power generation facilities based on operational characteristics?
- What characteristics distinguish between each cluster?
- Do the resulting clusters correspond to known fuel types despite fuel type not being included in the clustering algorithm?

Answering the above questions will allow us to develop awareness of the US power sector and show possible correlations between cost, efficiency, pollution, and energy output, as well as generalize information about the fuel sources used to power these US power plants.

Analysis and Discussion

A random sample comprising approximately 2% of the cleaned dataset was selected to reduce computational time while preserving representation of the overall data. Clustering variables included units of fuel received, fuel energy content (MMBtu per unit), fuel cost per MMBtu, total fuel received (MMBtu), total fuel cost, sulfur content, and ash content. These variables were standardized prior to clustering to prevent any single variable from controlling the analysis due to scale differences.

The elbow method and silhouette analysis indicated that five clusters offered an ideal balance between model complexity and cluster separation. Subsequently, a K-means model with five clusters was fitted to the standardized data.

The table below summarizes the standardized cluster centroids generated by the K-means algorithm. Cluster 2 is characterized by exceptionally high fuel receipt volumes, while Cluster 4 exhibits the highest energy content, total energy received, and sulfur and ash contents, indicative of large coal-fired generating facilities. Cluster 3 is an outlier due to its exceptionally high fuel cost per MMBtu, whereas Clusters 1 and 5 represent lower-volume and moderate-scale operating facilities, respectively.

1	-0.31	-0.49	0.02	-0.49	-0.43	-0.36	-0.40
2	2.05	-0.63	-0.08	0.81	1.25	-0.40	-0.40
3	-0.54	-0.63	35.55	-0.62	-0.57	-0.40	-0.40
4	-0.20	1.81	-0.12	3.07	2.00	1.66	1.30
5	-0.46	1.73	-0.12	0.22	-0.03	1.11	1.37

The resulting clusters can be summarized in more detail as follows:

- **Cluster 1** consisted primarily of lower-volume natural gas and oil facilities. These plants exhibited below-average fuel consumption, total energy received, and total fuel cost while continuing to have very low sulfur and ash content. This cluster represents smaller, cleaner-burning generating facilities, including many lower-utilization plants.
- **Cluster 2** represented large-scale natural gas facilities. These plants received substantially larger fuel quantities than average and incurred above-average total fuel costs, while producing very low sulfur and ash content. This cluster reflects high-output natural gas generating stations operating at a much larger scale than those in Cluster 1.
- **Cluster 3** contained only four observations and was defined by an exceptionally high fuel cost per MMBtu while all other variables remained below average.

Because of its extremely small size and unusual cost values, this cluster likely represents pricing anomalies or uncommon fuel purchase transactions rather than a meaningful operational category.

- **Cluster 4** consisted entirely of coal-fired facilities with the highest total energy received, above-average total fuel costs, and the highest sulfur and ash content among all clusters. These plants represent large baseload coal-generating stations with high fuel consumption and relatively high emissions.
- **Cluster 5** also contained coal-fired facilities but operated at a more moderate scale than Cluster 4. Although these plants exhibited elevated sulfur and ash content, their fuel consumption and total fuel costs were closer to the dataset average. This cluster represents medium-sized coal-generating facilities.

To further evaluate the clustering results, fuel type was compared with the cluster groupings, even though it was left out as a clustering variable. The comparison showed a strong relationship between the clusters and the underlying fuel categories. Natural gas and oil facilities were grouped into Clusters 1 and 2, while coal facilities were separated into Clusters 4 and 5. This indicates that the numerical variables selected for clustering appropriately captured differences in operational characteristics across fuel types.

1	0	2,577	800	3,377	Natural Gas / Oil
2	0	752	0	752	Natural Gas
3	0	4	0	4	Natural Gas (Outlier)
4	273	0	0	273	Coal
5	952	0	0	952	Coal
Total	1,225	3,333	800	5,358	

Overall, the analysis shows that K-means clustering successfully identified meaningful operational groups within U.S. power generation facilities based solely on numerical fuel characteristics. The resulting clusters distinguished plants by operational level, fuel quality, and environmental characteristics while also showing a small group of unusual fuel pricing observations that warrant additional investigation.

Conclusions

The analysis established that K-means clustering can reliably identify meaningful operational groups in U.S. power-generation fuel data. The resulting clusters reflected differences in plant scale, fuel quality, environmental characteristics, and fuel costs, closely corresponding to actual fuel types. These results indicate that numerical fuel characteristics are sufficient for classifying power plants without direct reliance on categorical fuel-type information.

Several assumptions were made during the analysis. Variables with substantial missing values were removed, remaining missing observations were omitted, and a two percent random sample was used to reduce computing requirements. Additionally, K-means assumes clusters are approximately spherical and is sensitive to extreme values, which likely contributed to the identification of the fuel cost outlier cluster.

Later investigations can look into alternative clustering methods, such as hierarchical clustering or DBSCAN, apply outlier detection prior to clustering, or incorporate additional plant characteristics to enhance cluster interpretation. Regardless of these limitations, the analysis shows that unsupervised machine learning is a valuable approach for identifying operational patterns in large energy datasets and gives insights into the diversity of fuel use across U.S. power plants.